

Kirti Pratihari

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GitHub: github.com/KirtiPratihari | LeetCode: leetcode.com/u/Kirti03/

Summary

Computer Science undergraduate specializing in AI & ML, with hands on experience developing Python based backend services, integrating external APIs, and building scalable, data driven applications. Proficient in Python, FastAPI, Docker, Git, and familiar with AWS and Microsoft Azure. Strong foundation in software engineering, system design, and problem solving, with an interest in cloud infrastructure, automation, and enterprise technology solutions.

Education

Vellore Institute of Technology, B.Tech in Computer Science (AI & Machine Learning) | CGPA: 8.85
Bhopal, India — Expected 2027

Technical Skills

- **Languages & Framework:** Python, C++, SQL, FastAPI, Rest API
- **Cloud & AI Ops:** Git, GitHub, Docker, AWS, Microsoft Azure (DP-900), Vercel, Render
- **Data & Vector Infrastructure:** Pinecone (Vector DB), Relational Databases, Data Pipeline Optimization, Semantic Indexing
- **AI & LLM:** LLMs & Embeddings, RAG Pipelines, Prompt Engineering, Agentic Workflows (LangChain, LangGraph), Neural Network

Projects

RAG Chat Assistant (YAP ENGINE)

[Live Link](#)

- Architected a modular Retrieval-Augmented Generation (RAG) system processing complex document structures using LangChain and Python.
- Optimized semantic indexing by deploying a high-density vector space pipeline in Pinecone, reducing retrieval latency for context-aware queries.
- Integrated the Groq API and engineered automated validation pipelines to rigorously evaluate AI generated outputs for correctness, consistency, and a reduction in hallucinations.

Credit Card Approval Prediction System

[GitHub Link](#)

- Developed an end-to-end predictive machine learning pipeline utilizing a Random Forest Classifier to achieve 90% accuracy.
- Addressed severe class imbalance by implementing SMOTE (Synthetic Minority Over-sampling Technique), boosting minority class recall and model fairness.
- Built a modular backend pipeline with automated feature extraction, schema validation, and real-time inference endpoints for financial demographic datasets

Taxi Trip Duration Prediction Analysis

[GitHub Link](#)

- Built a scalable regression pipeline using XGBoost, achieving a competitive 0.3151 Root Mean Squared Log Error (RMSLE) on urban transit datasets
- Engineered spatial-temporal feature transformations by applying K-Means clustering to coordinate data, mapping non-linear congestion corridors and traffic density
- Conducted rigorous exploratory data analysis and cross-validation to isolate predictive indicators that optimize routing and transit dispatch algorithms

Technical Experience & Leadership

Technical Lead, Matrix Club, VIT Bhopal University

Jan 2025 - Present

- Led technical operations for the 'Escape the Matrix' event, managing the digital registration infrastructure for 50+ participating teams.
- Served as the primary technical point of contact, ensuring seamless digital communication and support for club members.

Technical Training, India Space Lab Winter

Internship Dec 2024 - Jan 2025

- Executed advanced drone flight operations and analyzed space technologies including CubeSat integration and CanSat deployment.

Achievements & Certifications

- Awarded 1st Prize at InnovMinds Expo Hackathon (Feb 2025) for a dementia support app utilizing AI classification.
- Ranked 79 in India Space Lab Internship Exam (Merit-based national competition).
- **Microsoft Certified:** Azure Data Fundamentals (DP-900).
- Advanced Learning Algorithms - Stanford University (May 2025).
- Open-Source Contribution in Hacktober Fest